MILITARY SPECIFICATION

Igniter Mark 296 Mods (Loaded Assembly)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope This specification establishes the manufacturing and testing requirements for the procurement of Igniter Mark 296 Mods (Loaded Assembly), hereafter called the igniter.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications. standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment. General Specification for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspections by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardized Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-129	Marking for Shipment and Storage
MIL-STD-331	Fuze and Fuze Compponents, Environmental and Performance Tests for
MIL-STD-414 4	Sampling Procedures and Table for Inspection by Variables for Percent Defective
MIL-STD-810 o	Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command (Code Indent 30003)

1255AS114	Mark 296 Mod O Igniter. Explosive. Loaded Assembly
1255 AS133	Shipping Container, Mark 296 Mod O, Igniter
1255 AS141	Mark 296 Mods, Test Chamber Assembly
DL 1255 AS114	Igniter, Mark 296 Mod O, Loaded Assembly

PUBLICATIONS

Naval Sea Systems Command

OR 10	Documentation Requirements for Packaging, Packing, Shipping, and Handling of Weapon System Components
OR 11	Development of Packaging, Handling. Storage, and Transportation Systems for Weapons

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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Code of Federal Regulations

49 CFR 100-177 Transportation

(Copies are available from the Superintendent of Documents. Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite latest issue and supplements thereto.)

3. REQUIREMENTS.

3.1 Conformance with drawings. Unless otherwise specified in the contract, the igniter shall meet all the requirements specified herein and on the applicable drawings listed on Dl 1255 AS114.

3.2 First article. When specified in the contract (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4 and 6.4).

3.3 Circuit resistance. The igniter circuit resistance between the ends of the lead wires shall be between 0.40 and 0.60 ohms (see 4.6.2).

3.4 Performance requirements. The performance requirements of the igniter shall be as specified in Table I (see 4.6.10). O).

TABLE I. Performance requirements.

Requirement	0° F		+140° F	
	Min	Max	Min	Max
Initiator peak pressure (psia)	–	65	–	65
Pellet peak pressure (chamber psia)	95	210	100	220
Time to pellet peak chamber pressure (ms)	15	50	15	50

3.5 Environmental requirements.

3.5.1 Impact. The igniter shall withstand the 40-foot drop test without igniting (see 4.6.3).

3.5.2 Two-foot-drop. The igniter shall withstand the 2-foot drop test without spilling arty explosive material and without igniting (see 4.6.4).

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3.5.3 Temperature and humidity. The igniter shall withstand the temperature and humidity test (see 4.6.5).

3.54 Vibration. The igniter shall withstand the simulated shipboard vibration test (see 4.6.6)

3.5.5 Acceleration. The igniter shall withstand the 15 g load and acceleration test (see 4.6.7)

3.5.6 High temperature test. The igniter shall not ignite when subjected to a temperature of 350° F for 120 minutes minimum (See 4.6.8).

3.57 Salt Spray The igniter shall withstand the salt spray test without corrosion or damage (see 4.6 9)

3.5 8 T ransportation shock. storage, and handling. The igniters. when packed in accordance with Section 5. shall not rupture so as to spill the package contents or break the seals on unit packages (see 4.6. 11).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility y for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified in the contract (see 6.2.1), all inspections shall be performed in accordance with the test conditions specified herein.

4.4 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. Frost article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

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4.4.1 Sample size Thirty-two igniters shall be subjected to first article inspection.

4.42 inspection routine. The sample shall be subjected to all the inspections and tests of 4.6 in accordance with the testing schedule of figure 1.

4.4.3 Failure. Failure of any igniter in the sample to meet the above requirements shall be cause for refusal to grant first article approval.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of all the examinations and tests as noted in Table II. The minimum production inspection to be performed shall be in accordance with MIL-STD-105 105 and shall consist of all the requirements including the classification of defects listed on the applicable drawings. Acceptance shall be in accordance with 4.8.

TABLE II. Quality conformance inspection

Examination	Requirement	Test
Visual examination	3.1	4.7.1
Circuit resistance	3.3	4.7.2
Performance test	3.4	4.7.3

4.5.1 Inspection lot. An inspection lot shall consist of igniters manufactured from one batch of main charge pellets and one lot of initiators.

4.5.2 Sampling.

4.5.2.1 Circuit resistance sampling All igniters in a lot shall be tested for conformance to the circuit resistance requirements (see 3.3).

4.5.2.2 Performance requirements sampling. Sampling of loaded assembly igniters for conformance to the performance requirements in 3.4 for each temperature shall be in accordance with MIL-STD-414 14, Table A2, Inspection Level 111. Inspection Level 11 shall only be used after 3 consecutive accepted lots have been manufactured. If a lot fails, Level III once again shall be in effect.

46 First article tests,

4.6.1 Visual examination. The igniters shall be examined for conformance to 3.1.

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4.6.2 Circuit resistance. The igniter circuit shall be checked for circuit resistance in accordance with the requirements of 3.3. Not more than 0.005 ampere shall be applied to the igniter circuit during the test.

4.6.3 Impact test. Two igniters shall be subjected to a 40-foot drop test in accordance with MIL-STD-331. Igniters shall not fire during the test and shall be safe to handle and dispose after the test. Igniters do not have to be usable after the test.

4.6.4 Two-foot drop test. Three igniters shall be subjected to a 2-foot drop test along the longitudinal axis. nose down, at ambient temperature. Acceptance shall be in accordance with 4.6.10.

4.6.5 Temperature and humidity. Five igniters shall be subjected to the temperature and humidity test. Igniters shall be subjected to alternate temperature conditions of 0° F and +140″ F; time at each temperature will be a minimum of 18 hours with a 5 minute maximum transfer time. Three full cycles will be required. A full cycle is defined as 0° to + 140° to 0° F with relative humidity (RH) equal to 95 percent. After completing the test, two of the igniters shall be test fired at 0° F and two at +1 40° F in accordance with the testing schedule of Figure 1. Acceptance shall be m accordance with 4.6.10.

4.6.6 Vibration. Five igniters shall be vibrated along the longitudinal and transverse axes. Igniters shall be mounted in a test vehicle which simulates the mounting of the igniter _ in the Rocket Motor Mark 23 Mods. Test Procedures and time schedules shall be in accordance with MIL-STD-810 10. figure 514-6, curve AA, beginning at 5Hz. After the test, two of the igniters shall be test fired at 0° F and two at + 140° F m accordance with the testing schedule of Figure 1. Acceptance shall be in accordance with 4.6.10.

4.6.7 Acceleration. Five igniters shall be mounted simulating the actual mounting of the Rocket Motor Mark 23 Mods and subjected to 15 ± 2 g acceleration load. The test shall be repeated 10 times. The g load shall be reached in 7 ± 1 milliseconds and the acceleration shall exceed 8 g for a minimum of 11 milliseconds. The igniters shall be in such a position that the load acts along the longitudinal axis of the igniters. After the load and acceleration test, two of the igniters shall be test fired at 0° F and two at + 140° F in accordance with the schedule of Figure 1. Acceptance shall be in accordance with 4.6.10.

4.6.8 High temperature test. Four igniters, in accordance with testing schedule of Figure 1, shall be subjected to high temperature test. The igniter shall be enclosed in a test chamber, the temperature of which shall be 350° ± 20° F. After the igniter has been subjected to this temperature for the specified minimum time of 2 hours, it shall be test fired in accordance with the testing schedule of Figure 1.

4.6,9 Salt spray test. Four igniters shall be loaded into a simulated Rocket Motor Mark 23 and subjected to the salt spray test in accordance with MIL-E-5272. 272. After completing the test, two of the igniters shall be test fired at 0° F, and two at + 140° F in accordance with the testing schedule of Figure 1. Acceptance of igniters shall be in accordance with 46 10.

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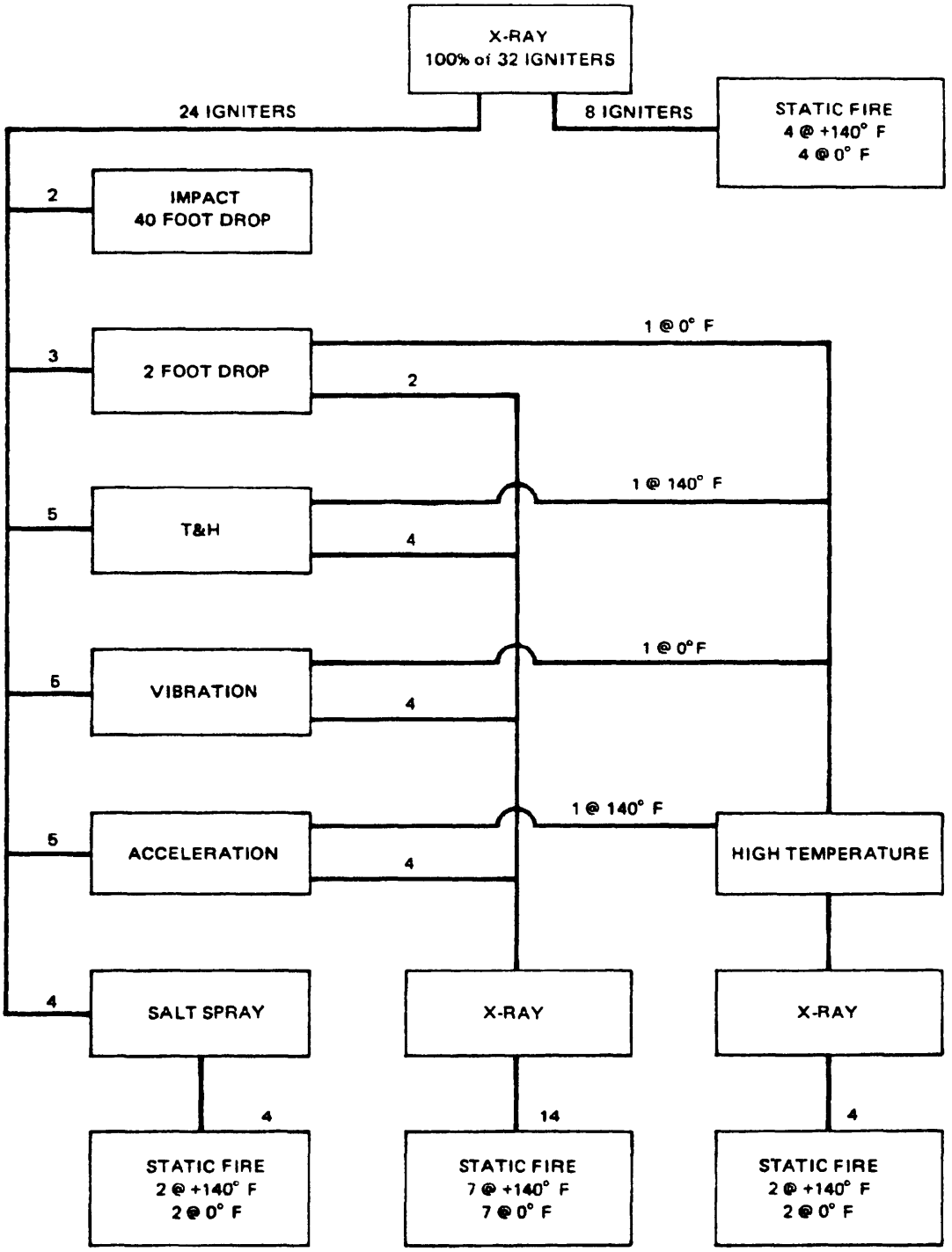


FIGURE 1. Testing schedule for first article sample.

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4 6b 10 First article performance test. Fifteen igniters shall be test fired at 0" F and fifteen teen at +140" F in accordance with the testing schedule of Figure 1. The requirements of 3.4 shall be evaluated at each temperature in accordance with Table B-3 of MIL-STD-414 using an AQL of 0.65 percent.

4.6.11 Transportation, shock, storage, and handling. Three igniter packs, prepared as specified in Section 5, shall be subjected to the appropriate rough handling tests specified in OR 10 or OR 11.

4.7 Quality conformance tests.

4.7.1 Visual examination test. The igniters shall be examined for conformance to 3.1.

4.7.2 Circuit resistance test. After the completion of the igniter assembly, the igniter circuit shall be checked for circuit resistance in accordance with the requirements of 3.3. Not more than 0.005 ampere shall be applied to the igniter circuit during the test.

4.7.3 Performante test.

4.7.3,1 Test equipment required. The following test equipment shall be used for performance tests:

a. Test chamber shall be in accordance with Drawing 1255AS141. Condition the test chamber to 70° ± 35° F before firing the igniter.

b. Reluctance type pressure pickups with 8-inch maximum length, 1/4-inch inside diameter air filled line for connection to test chamber.

c. Recording oscillograph, Type 5-114, Consolidated Engineering Corporation, Pasadena, California, or equal.

d. Necessary couplings.

e. Twenty-eight volt d .c. power supply.

4,7.3.2 Temperature conditioning. The igniters selected in 4.5.2.2 shall be conditioned to the applicable temperature for a minimum of 6 hours. Igniters shall be fried within 5 minutes after removal from the conditioning unit. If they are not fired within this time, the y shall be returned for reconditioning for an additional 6 hours.

4.7.3.3 Power supply. Igniters shall be fired using a" 28-volt power supply and current of 13 ± 2 amperes as delivered to the igniter circuit.

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4.7.3.4 Recorded data A pressure time curve shall be reduced from the oscillograph records, and a permanent record made of the parameters specified in 3.4. All time measurements shall be made at, or referred to, the instant the electrical firing circuit is closed. Ambient prefiring pressure shall be the reference datum for all pressure measurements.

4.8. Acceptance criteria. The failure to meet the performance requirements due to a malfunction attributed directly to the test apparatus shall not be cause for rejection of the lot. A new sample shall be drawn and the test rerun. A malfunction of the igniter hardware that occurs during the performance tests shall be reported to the procuring activity with details of the malfunction.

4.8.1 Production acceptance.

4.8.1.1 General and circuit measurements. The acceptable quality level (AQL) for general examination acceptance shall be as follows.

a. Critical	100 percent inspection
b. Major	AQL 1.00 percent
c. Minor	AQL 2.50 percent

4.8.1.2 Performance requirements. The test results shall meet the requirements of 3.4 with samples selected in accordance with MIL-STD-414 14, Table B-3.

5. PACKAGING

51 Preservation and packaging.

5.1.1 Level A. In addition to the following, Level A packaging shall be in accordance with 49 CFR 171-177.

5.1.1.1 Cleaning, drying, and preservative application. Cleaning, drying, and preservative application shall be in accordance with applicable drawings and specifications listed m DL 1255A1 14.

5.1.1.2 Unit packaging. Unit packaging of the igniters shall be in accordance with Drawing 1255AS1 33.

5.2 Packing.

5.2.1 Level A. In addition to the following, Level A packing shall be in accordance with 49 CFR 171-177.

5.2.2 Exterior containers. Unless otherwise specified in the contract or purchase order, eight loaded assembly igniters packaged m accordance with 5.1.1.2 shall be packed in accordance with Drawing 1255AS1 33.

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5.3 Marking.

5.3.1 Special marking. Marking of exterior containers shall be in accordance with the 49 CFR 171-177.

5.3.2 Normal marking. In addition to the markings required by contract or order, unit packages and shipping containers shall be marked m accordance with the requirements of MIL-STD-1 29.

6 NOTES

6.1 Intended use. The Igniter Mark 296 Mods (Loaded Assembly) is intended for use in the Rocket Motor Mark 23 Mods 2 and 3.

6.2 Ordering data.

6.2.1 Acquisition requirements Procurement documents should specify the following:

a. The title, number, and date of this specification

b. If first article is required and, if so, sample size

c. Activity to inspect and test the first article sample

d. Sampling, if different than in 4.5.2

e. Distribution of performance

f. All applicable drawings and latest revisions

g. Packaging and marking, if different than specified herein (see 5.2 and 5.3)

h. Inspection conditions if other than specified herein (see 4.3)

i. That the safety precaution requirements of the Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material, DOD 4 1 45 .26M, are applicable. NOTE. When this specification is used as part of the description of work to be accomplished by the Government Activity, the safety precaution requirements of Ammunition Ashore. OP5, are applicable.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into

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the contract. When the provisions of DAR 7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited m the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID No.	Option
4.7.3.4	Oscillograph Instrumentation Data Record	D1-T-1915	
4.8	Report, Failure Summary and Analysis	D 1 -R-7041	

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000. 19 L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions

6.3.1 Initiator peak pressure. The maximum pressure achieved by the initiator charge.

6.3.2 Pellet peak chamber pressure. The maximum pressure achieved by the igniter main charge.

6.3.3 Time to pellet peak chamber pressure. The elapsed time between energizing the igniter and the attainment of pellet peak chamber pressure.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations. The first article should consist of 32 units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.5 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein should be referred in writing to the procuring activity or appointed agent for interpretation and clarification.

6.6 Requests for deviation. Requests for deviation from this specification, applicable drawings or any specification, publication, material, and process specified herein should be forwarded to the procuring activity prior to manufacture of the affected item or items. Al! deviations should be limited to the contract under which they were granted.

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6.7 Requests for waivers. Requests for waiver from this specification, applicable drawings. specifications, materials. or process should be forwarded to the procuring activity. A waiver request should include the following:

a. Exact nature of defect
b. Drawing number and defect classification
c. Quantity of items involved.

6,8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity:
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(Project 1340-N630)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-I-85365A(AS) Igniter Mark 296 Mods-(Loaded Assembly)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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